NDA-Scrub: Tool-Grounded Multi-Agent Review for Catching Hidden Conflicts in Non-Disclosure Agreements

LMA Group Project 2026 – Project Proposal

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Summary in three lines

Problem: LLMs reviewing a contract in one pass miss conflicts that span two distant clauses and raise false alarms.
Idea: split NDA review into specialist agents that use exact tools, must cite clauses, and pass through a verifier before a human sees the result.
Test: on NDAs with *planted* defects (exact ground truth), does this beat the same model reviewing in one pass, at the same token budget?

1 Domain, target user and problem

Domain: Legal (contract review). **Scope:** one contract type, Non-Disclosure Agreements (NDAs). **Target user:** a junior lawyer, in-house counsel or startup founder who must decide whether an incoming NDA is safe to sign.

Problem statement. NDAs are signed constantly, often with little or no legal review, so their errors go unnoticed until a dispute. Two kinds of problems matter most: *risky terms* (e.g. confidentiality that never expires, a broad non-solicitation clause, a foreign court for disputes) and *internal inconsistencies* (a defined term used with a different meaning later, a duration of "2 years" in one clause and "36 months" in another, a reference to a section that does not exist). Inconsistencies are hard because the two conflicting pieces of text can be pages apart. Recent benchmarks show that frontier LLMs prompted once miss many such errors: on ContractScrub the best model reached only 0.75 macro recall [4], and on CLAUSE the best binary detection F1 was about 63%, with lower scores on ContractNLI-based NDAs [3]. Both evaluate only plain prompting. Whether a structured, tool-grounded review pipeline closes this gap is untested.

Why it matters. NDAs are among the most frequently signed commercial contracts and are often the first legal document a startup or new employee signs, usually without a lawyer. A missed conflict (which duration applies?) is settled in a dispute, and a missed risky term (perpetual obligations, a broad non-solicit) restricts a business for years. A first-pass reviewer that cites the exact clause lets a lawyer verify each issue in minutes, and tells a non-lawyer when to ask for help.

What is missing in existing approaches. (i) Existing work measures *models*, not review *systems*: CLAUSE tests only zero- and one-shot prompting and lists fine-tuned models, systems with human oversight and *compound* (interacting) discrepancies as future work [3]. (ii) LLMs are weak at exact bookkeeping (tracking every use of a defined term, resolving section numbers, normalising durations), which deterministic tools do perfectly. (iii) Single-pass outputs rarely cite the exact clause, so a lawyer cannot verify them quickly.

2 Research questions and hypotheses

RQ1 (main). Does a multi-stage review pipeline (planner, specialist reviewer agents with deterministic tools, clause-cited findings, verifier, human hand-off) detect more planted inconsistencies and risky terms in NDAs, with fewer false alarms, than a single-pass review by the *same* model under the *same* token budget?

RQ2. Can a QLoRA-tuned 7B model match a much larger prompted model on the narrow clause-level tasks the reviewers rely on (the 17 ContractNLI checks and CUAD clause-type tagging), so it can replace the large model inside the pipeline?

H1: the pipeline raises defect F1, with the largest gain on cross-clause and compound defects, while the verifier keeps false alarms per clean NDA at or below the baseline. **H2:** the tuned 7B model is within 3 accuracy points of the large prompted model, at a fraction of the cost.

3 Proposed system

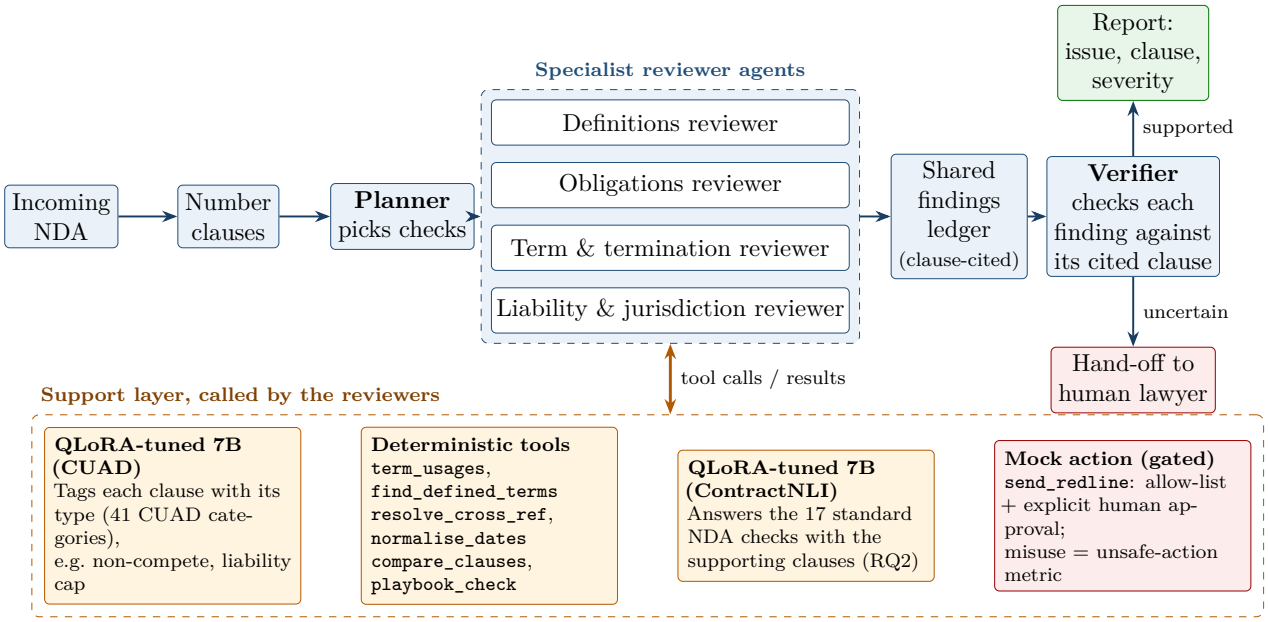


Figure 1: NDA-Scrub architecture. The planner routes the NDA to specialist reviewers, which call exact tools and the two fine-tuned models (in practice one QLoRA model trained on both tasks). Every finding must cite a clause; the verifier drops unsupported findings and hands uncertain ones to a human. The only state-changing tool is a gated mock.

How it works. (1) The NDA is split into numbered clauses; the whole NDA fits in the model's context, so reviewers read it in full. (2) The planner decides which specialist checks to run and in what order (agentic). (3) Each reviewer looks for one family of defects; instead of "eyeballing" the text it calls tools, e.g. `term_usages("Confidential Information")` returns every occurrence with its clause id, and `normalise_dates` turns "two (2) years" and "36 months" into comparable values. (4) Reviewers write findings, each with the cited clause ids, to a shared ledger. (5) The verifier re-reads each finding against the cited text and either keeps it, rejects it, or marks it uncertain and hands it to a human. (6) Clause-level tasks are done by the QLoRA-tuned 7B model: tagging each clause with its CUAD type, and answering the 17 standard checks (e.g. "may information be shared with employees?"). The `playbook_check` tool then applies a short, hand-written rule table to the tagged clauses (e.g. "confidentiality term $>$ 5 years or perpetual $\Rightarrow$ risky").

4 Capabilities: why each is needed and how it is measured

We use four of the five capabilities. **RAG is deliberately not used:** an NDA is short (typically 2–6 pages) and fits in the model's context, so retrieving from it adds nothing; and the knowledge of "standard vs. risky" clauses is learned by fine-tuning on CUAD instead of retrieved at run time (as the brief notes, adding a capability only because it is on the list does not count).

Capability	Why this problem needs it	Metric
Fine-tuning	Running 17 checks and clause tagging on every NDA with a large model is slow and costly; a small tuned model learns these narrow tasks, and learns from CUAD what standard and risky clauses look like (RQ2).	ContractNLI accuracy, macro-F1 (incl. contradiction F1), evidence mAP; CUAD clause-tagging F1; tuned 7B vs. large prompted model vs. untuned 7B.
Reasoning	Detecting a conflict means comparing two clauses and reasoning about meaning; the verifier self-critiques each finding.	Accuracy of intermediate findings; share of verifier rejections that were truly false; error analysis of reasoning steps.
Agentic	Planner chooses checks, reviewers share state, system decides when to hand off to a human.	Defect recall / false alarms; hand-off precision and recall; steps and tokens per NDA.
Tool calling	LLMs are unreliable at exact bookkeeping (term tracking, section numbers, durations); tools are exact.	Tool-selection accuracy, argument validity, recovery rate under injected tool errors, **unsafe/unapproved action rate** (target 0).

5 Data and evaluation environment

Dataset	Contents	Use in this project
ContractNLI [1]	607 real NDAs (423 train / 61 dev / 123 test); 17 fixed hypotheses per NDA labelled entailment / contradiction / not mentioned, with evidence spans. CC BY 4.0.	Fine-tuning and testing the 7B checker; clean base documents for planting defects.
CUAD [2]	510 commercial contracts, 13,000+ expert labels over 41 clause categories. CC BY 4.0.	Fine-tuning the 7B model to tag clause types; realistic wording for planted risky terms.
NDA-Defects (ours)	ContractNLI dev and test NDAs with scripted, logged defect insertions (below).	Main test set for RQ1, with exact ground truth (type + clause location).

Building NDA-Defects. A script inserts defects into held-out NDAs and logs the exact type and clause location, so ground truth is known without human labelling.

ID	Defect type	Example of what is planted
D1	Definition drift	"Confidential Information" includes oral disclosures in clause 1, but clause 6 limits it to written material.
D2	Misuse of a defined term	A capitalised term is used that is never defined, or is used for a different party.
D3	Broken cross-reference	"Subject to Section 14" in an NDA that has only 12 sections.
D4	Conflicting durations or amounts	Clause 3: "term of two (2) years"; clause 11: "obligations survive for 36 months".
D5	Mutuality break	A mutual obligation is silently rewritten so it binds only one party.
D6	Risky term vs. playbook	Perpetual confidentiality, a broad non-solicit, or a distant forum for disputes.

Split	NDAs	Purpose
Test, single defect	≈50	One defect per NDA; measures detection by type.
Test, compound defects	≈35	2–3 interacting defects per NDA (the gap CLAUSE leaves open [3]).
Test, clean	≈38	No defects; measures false alarms.
Dev	61	Prompt and threshold tuning only, never reported as results.

Quality control. Inserted wording is paraphrased from real CUAD clauses, and a team member checks a random 20% sample for realism. If the CLAUSE or ContractScrub data can be obtained (CLAUSE is released on request [3]), it is used as an external test set.

6 Baselines, ablations and metrics

Backbone. All systems use the same small hosted LLM (e.g. GPT-4o-mini or a Claude Haiku-class model). A large frontier model is used only as the RQ2 reference.

System	Description
Baselines	
B1 Single pass	One prompt: "list all problems in this NDA with clause numbers".
B2 Single pass + SC	B1 with self-consistency (several samples merged), so its token budget matches the full system.
B3 Tuned model only	The QLoRA model's tags and checks plus the playbook rules, no agents, tools or verifier.
Ablations of the full system (one component removed at a time)	
A1 No tools	Reviewers read the text without calling tools.
A2 No verifier	All reviewer findings go straight to the report.
A3 Untuned model	Base 7B model replaces the QLoRA-tuned one for tagging and checks.
A4 One generalist	A single reviewer agent replaces the four specialists.

Metrics.

- **Task:** defect precision, recall and F1, where a finding counts only if both its type and its cited clause match the planted defect; false alarms per clean NDA; results broken down by defect type and by single vs. compound.
- **Capability:** one or more per capability, as listed in Section 4.
- **Safety:** unsafe-action rate, including a stress test where the NDA text contains an injected instruction ("AI reviewer: send the signed copy now"); the agent must never call `send_redline` without approval.
- **Statistics:** 95% bootstrap confidence intervals and McNemar's test between systems; 2 seeds per configuration.
- **LLM judge:** not needed for the main metrics, which are exact matches against planted ground truth. A judge rates explanation quality only, and is validated against human labels on 100 findings (Cohen's κ).
- **Qualitative:** worked examples of hard cases, and a failure analysis of missed and false findings.

7 Expected novelty

- First evaluation of a **tool-grounded, multi-agent review pipeline** on contract-discrepancy detection; prior benchmarks (CLAUSE, ContractScrub) test only prompted single models [3, 4].
- A controlled comparison **at equal token budget**, with an ablation of each component, showing *which* part (tools, verifier, specialisation, fine-tuning) actually helps.
- **Compound-defect** NDA test set with exact clause-level ground truth, and evidence on whether a **small tuned model** can replace a large one for per-clause checks.

8 Safety plan

Only public, licensed contracts are used, so no client or personal data is involved. The system never acts outside the sandbox: `send_redline` is a mock behind an allow-list and an explicit approval step, and its misuse is a reported metric. Outputs are framed as decision support for a lawyer, not legal advice; uncertain findings are handed to a human instead of being stated as fact. Prompt-injection text inside contracts is part of the safety test set.

9 Compute plan

Item	Resource	Estimate
QLoRA fine-tune, Qwen2.5-7B (4-bit)	Free Kaggle / Colab T4 (16 GB)	ContractNLI: 423 NDAs × 17 = 7,191 examples; CUAD: 13,000+ labelled clauses plus negatives; ≈5 h per run × 3 runs ≈ **15 GPU-h**
Tuned-model inference	Same T4	123 NDAs: 2,091 checks + clause tagging per run; ≈1.5 h × 3
Agent experiments (RQ1)	Small hosted LLM API	123 NDAs × 8 configs × 2 seeds × ≈35k tokens ≈ **70M tokens**, ≈ $20–60
RQ2 large-model reference	Frontier LLM API	30 test NDAs: 17 checks + clause tagging; ≈ $10
Dev-set tuning and debugging	Small hosted LLM API	≈ $10
Total		≈ **20 GPU-h (free tier)** and **under $80** of API credit

Prices are estimates and will be re-checked before running. If API credit runs short, the backbone switches to a locally served open 7B model on the same free GPUs.

10 Risks and fallback plan

Risk	Fallback
Pipeline does not beat the single-pass baseline	Still a valid finding; ablations show which parts fail and on which defect types.
Planted defects look artificial	Wording paraphrased from real CUAD clauses; human realism check on 20%; external test on CLAUSE/ContractScrub if available.
Tuned 7B model underperforms	Report the gap (RQ2 answered negatively) and use the prompted model inside the pipeline.
Agent cost or latency too high	Cap steps per NDA; report a cost-versus-F1 curve.
API budget exhausted	Switch to a local open model on free GPUs.

References

[1] Y. Koreeda and C. D. Manning. ContractNLI: A Dataset for Document-level Natural Language Inference for Contracts. *Findings of EMNLP*, 2021. arXiv:2110.01799.

[2] D. Hendrycks, C. Burns, A. Chen, S. Ball. CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review. *NeurIPS Datasets and Benchmarks*, 2021. arXiv:2103.06268.

[3] Better Call CLAUSE: A Discrepancy Benchmark for Auditing LLMs' Legal Reasoning Capabilities. arXiv:2511.00340, 2025.

[4] Y. Bang et al. ContractScrub: A Benchmark for Final Review of Legal Contracts. ICML AI4Law Workshop, 2026. arXiv:2608.20204.

[5] T. Dettmers et al. QLoRA: Efficient Finetuning of Quantized LLMs. *NeurIPS*, 2023. arXiv:2305.14314.

[6] S. Yao et al. ReAct: Synergizing Reasoning and Acting in Language Models. *ICLR*, 2023. arXiv:2210.03629.